Optimizing Transmon Qubit Readout Time via Signal-to-Noise Ratio Characterization

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High-fidelity dispersive readout requires balancing signal accumulation against qubit relaxation (T_1). We characterize this trade-off by implementing an automated QUA node to sweep integration time on a fixed-frequency transmon qubit. By decomposing the measurement record into signal and noise contributions, we experimentally validate the fundamental dispersive noise model, observing the characteristic $\propto \sqrt{t}$ scaling of the signal-to-noise ratio. Incorporating qubit relaxation during readout, we identify an optimal integration time of $t_{opt} \approx 9.3$ μs that maximizes total measurement fidelity. Implementing this optimized parameter reduces readout latency by 42% compared to the baseline calibration while maintaining a peak fidelity of 77.3%. These results demonstrate that for systems with moderate signal-to-noise ratio, optimizing the temporal integration window is a critical step toward minimizing latency in quantum error correction cycles.

I. INTRODUCTION AND THEORY

High-fidelity and low-latency qubit readout are essential factors for quantum control, calibration, and error correction. Superconducting circuit QED (cQED) systems operate using transmon qubits that are measured dispersively. In this process, a readout resonator acquires a frequency shift dependent on the qubit state, and the driving frequency ω_d, with a microwave pulse, produces a complex amplitude in the complex I/Q plane that reveals the qubit state. In this paper, we adopt the standard dispersive measurement theory, which includes the appearance of state-dependent pointer states and their separation in phase space, this being the central theme of cQED measurement theory, as discussed in Girvin's notes on cQED at Les Houches (Chapters 6-7).

A. Dispersive Pointer Separation

During the readout, the amplitude of the resonator evolves to one of the following states (corresponding to the ground $|g\rangle$ and excited $|e\rangle$ states):

$$\alpha_g(t), \quad \alpha_e(t).$$

With that in mind, it is important to consider the separation in phase space between the two amplitudes:

$$\Delta\alpha_(t) = \alpha_e(t) \ - \alpha_g(t) \tag{1}$$

which determines how distinguishable the two measurement outcomes are.

Now, in the dispersive steady state (where t >> $1/\kappa$), this separation saturates to a constant maximum value, which we denote as $\Delta\alpha_{ss}$. Thus, let $\Delta\alpha(t) \to \Delta\alpha_{ss}$.

Continuing, we must take into account the parameters that determine steady-state separation in the dispersive regime. These parameters are resonator pull 2χ, resonator linewidth κ, and readout drive amplitude, as described in Girvin's Circuit QED Les Houches notes, Chapter 7, especially Section 7.2 on dispersive readout. However, our experiment does not tune these parameters since they are used as fixed properties of the system.

B. Integrated SNR Scaling

Let us now use the fact that in the presence of an amplifier and vacuum noise, the measurement record is corrupted by Gaussian noise with variance σ^2. Using the scaling implied by the integrated homodyne current in [1] (Eqs. 5.3–5.5), signal-to-noise ratio grows as the square root of the measurement duration:

$$\mathrm{SNR}(t) \approx \frac{|\Delta\alpha_{ss}|}{\sigma}\sqrt{t} \tag{2}$$

This is the main expression guiding our optimization task. Notice that it assumes that the resonator has already "rung up" and is in the steady state, i.e. t >> $1/\kappa$. In our case, our sweep starts at 100 ns, comparable to a few $1/\kappa$ for typical κ in the MHz range, so we neglect the initial transient response for this analysis. This expression also shows the fact that noise averages down as $\propto 1/\sqrt{t}$, while the pointer separation remains constant (after a short transient). The Girvin notes emphasize this same scaling when discussing homodyne detection of coherent states (see chapter 7 section 2.1).

C. Assignment Fidelity (Without T_1)

Now, if only Gaussian noise were present, the distinguishability of two Gaussian clouds leads to the well-known assignment fidelity:

$$F(t) \approx \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{\mathrm{SNR}(t)}{2\sqrt{2}}\right)\right] \tag{3}$$

This follows from the standard Gaussian discrimination model discussed in Gambetta's 2008 analysis [1] and from the standard result for discriminating two Gaussian distributions with equal variance, where the classification error is expressed in terms of the error function, which is consistent with the Gaussian pointer–state picture discussed in the Girvin Les Houches notes.

D. Including Qubit Relaxation During Readout

The readout is not instantaneous, which allows a qubit in an excited state to decay with a characteristic time T_1 during the integration window. As emphasized in the cQED literature (e.g., Walter et al. 2017), assignment fidelity and decay fidelity are typically separated conceptually:

Assignment fidelity: measures the fidelity of the separation of the two Gaussian probability distributions corresponding to the ground and excited states respectively (noise-limited).

Decay fidelity: measures the probability that the qubit has not died during the readout (lifetime-limited).

Thus, we approximate the total measurement fidelity as the product of separation fidelity and the probability of no decay:

$$F_{\text{measure}}(t) \approx F(t)\, e^{-t/T_1} \tag{4}$$

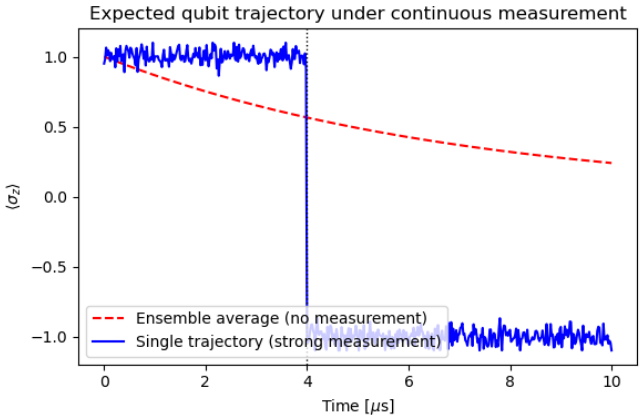


FIG. 1. **Visualization of readout-induced decay.** A simulated single-shot quantum trajectory (blue solid line) illustrating a quantum jump from $|e\rangle$ to $|g\rangle$ at $t \approx 4$ μs, compared to the ensemble average decay e^{-t/T_1} (red dashed line). This demonstrates the mechanism limiting readout fidelity: if the integration time t exceeds the jump time, the measurement is corrupted.

This phenomenological model is a heuristic approximation that captures the competing effects of noise-limited Gaussian separation and T_1 limited relaxation, and is commonly used in simplified analyses of dispersive readout. From this expression it can be noticed that as $t \to \infty$, higher SNR implies better Gaussian separation, but also, as $t \to \infty$, there is a higher chance of qubit decay and the measurement fidelity reduces. Therefore, since these two mechanisms oppose each other, the total fidelity features a single global optimum.

E. Fidelity and Optimal Time

Our project implements an automated QUA node that measures the I (In-phase component) and Q (Quadrature component) distributions as a function of integration time. Then, it extracts the magnitude of the separation of the two distribution amplitudes $|\Delta\alpha|$ from Eq. (1), it also extracts the standard deviation σ, and the SNR from Eq. (2). Next, the node proceeds to compute the fidelity curve as:

$$F_{\text{measure}}(t) \approx \frac{1}{2}\left[1 + \text{erf}\left(\frac{\text{SNR}(t)}{2\sqrt{2}}\right)\right] e^{-t/T_1}, \tag{5}$$

From that expression, the node identifies the optimal readout duration,

$$t_{\text{optimal}} = \arg\max\, F_{\text{measure}}(t).$$

This optimal time is found by evaluating $F_{measure}(t)$ at discrete sweep points. Then is extracted from the obtained output and is suggested to be the new calibration value for this parameter for the Qolab stack, allowing all future experiments to benefit from maximized readout fidelity.

II. METHODS

In this section, we describe the experimental setup and complete workflow used to measure readout fidelity as a function of integration time. This includes the hardware environment, essential calibration routines, the QUA node structure we designed, and the data processing process used to extract the signal-to-noise ratio (SNR) and measurement fidelity.

A. Hardware Configuration

The experiment is performed on a single fixed-frequency transmon qubit that is dispersively coupled to a readout resonator. Control and measurement are implemented with the Quantum Orchestration Platform (QOP), which provides us with precise waveform generation, timing, and real-time digitization using QUA programs. Assuming that the standard cQED measurement chain is the same architecture described in Girvin's Circuit QED Les Houches notes, which is widely used in transmon experiments, it follows that during readout, the outgoing microwave pulse is amplified by a cryogenic parametric amplifier, followed by a HEMT amplifier at 4 K, and additional room-temperature gain.

Subsequently, the readout pulse is downconverted and digitized, and the measure and acquire QUA commands are used to stream the I (In-phase) and Q (Quadrature) samples to the host computer.

Usually, these experiments are performed at a dilution refrigerator base temperature of approximately 10 mK.

B. Calibration Prerequisites

Several steps must be completed before sweeping the integration time:

Qubit drive calibration. Rabi and Ramsey experiments are used to determine accurate π and $\pi/2$ pulse amplitudes so the qubit can be reliably prepared in the ground and excited states.

Readout frequency calibration. Resonator spectroscopy is used to locate the probe frequency that maximizes the separation of the pointer states α_g and α_e.

Readout amplitude tuning. The readout drive amplitude is chosen to achieve good separation of the two steady-state coherent states without causing resonator nonlinearity.

Mixer calibration. Minimized IQ offsets and sideband leakage in order to reduce distortions in the Gaussian measurement distributions.

Timing Alignment. The digitizer integration window is aligned with the incoming measurement pulse so that the recorded amplitudes correspond accurately to the theoretical $\alpha_g(t)$ and $\alpha_e(t)$ defined earlier.

T1 time. The T1 time must be calculated to the most recent calibration of each qubit using the Qolab $05/_T1module$

These calibration steps are the most commonly used in standard cQED readout experiments and follow procedures described in the literature, including Gambetta et al. [1] and Walter et al. [2].

C. QUA Node: Integration-Time Sweep

The main contribution of our project is the implementation of a QUA node that measures SNR and fidelity across different integration times.

A continuous measurement of the state over time t follows the pattern below:

1. State preparation. The qubit is prepared alternately in $|0\rangle$ and $|1\rangle$ (corresponding to $|g\rangle$ and $|e\rangle$).

2. Readout pulse. A microwave readout pulse with duration t is applied.

3. Data acquisition. QUA's streaming mechanism collects the I and Q samples for each shot.

4. Averaging. The average I and Q values are extracted from the samples.

The integration times range from 100 ns to 16,100 ns, using 200 ns intervals for $|0\rangle$ and $|1\rangle$ respectively. The integration time was derived from the provided integration time within the Qolab stack of 4000 clock cycles, equivalently 16,000 ns. Each individual shot lasts roughly 16 µs (including readout and qubit reset), so the sweep takes about 1 ms of device runtime per qubit.

D. Data Processing Pipeline

Once the QUA node collects the raw I/Q data, the host computer processes the results through the following steps:

1. Histogram construction. For each integration time and each prepared state, the I/Q samples are separated into two datasets: one for $|0\rangle$ and one for $|1\rangle$. Each dataset forms a Gaussian-shaped distribution in the I/Q plane, which is consistent with the standard noise model in the reference literature.

2. Gaussian fitting. From each Gaussian distribution, the mean amplitude for the ground-state cloud (α_g), the mean amplitude for the excited-state cloud (α_e), and the noise standard deviation σ are extracted.

Then, using Eq. (1), the magnitude of the pointer separation $|\Delta\alpha|$ is computed.

3. SNR computation. The SNR is calculated for every integration duration using the integrated SNR definition from Eq. (2).

4. Fidelity computation. The measurement fidelity is computed using the model in Eq. (5), which combines the noise-limited assignment fidelity with the T_1 limited decay fidelity.

5. Optimal-time extraction. The integration time that maximizes the fidelity curve is identified as:

$$t_{\text{optimal}} = \arg\max F_{\text{measure}}(t).$$

This value is then converted to QUA clock cycle units and proposed as the new readout length parameter in the Qolab stack.

III. RESULTS

Figure 2 shows the calculated SNR derived from Equation (2) with a plotted line at the optimally calculated time, derived in Figure 3. The SNR value was 2.37 at that time, displaying the characteristic $\sqrt{t}$ scaling predicted by dispersive readout theory. The sublinear growth of SNR demonstrates the fundamental noise-averaging mechanism: while the integrated signal accumulates linearly with time, quantum and amplifier noise average down as $1/\sqrt{N_{\text{samples}}} \propto 1/\sqrt{t}$, resulting in the observed square-root improvement. The curvature of the SNR reveals diminishing returns at longer integration times. Early in the integration window (100-2000 ns), SNR grows rapidly as the measurement benefits significantly from additional averaging. However, beyond 5000 ns, the slope noticeably decreases, indicating that further integration provides progressively smaller SNR improvements. This saturation behavior is critical for understanding the optimization: at 9300 ns, the incremental SNR gain from longer integration becomes insufficient to justify the increased risk of T_1 decay.

Figure 3 displays the result of fidelity measurement as modeled with both assignment fidelity, from Equation (3), and measurement fidelity provided by Equation (5). This qubit was chosen due to its stable T_1 decay of 70.504 μs, as measured using the Qolab module 05_T1. Given a T_1 value within this range, the bounding of measurement fidelity exhibits distinct behavior compared to typical transmon qubits. The blue dashed curve representing

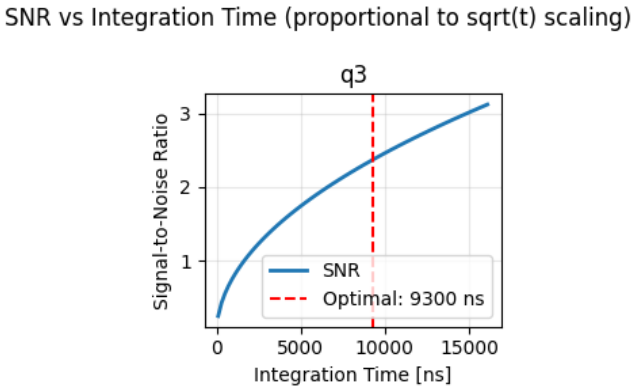


FIG. 2. Signal-to-noise ratio versus integration time, demonstrating $\sqrt{t}$ scaling as predicted by Equation (2). The red dashed line marks the optimal integration time of 9300 ns, where SNR = 2.37. The concave shape reflects diminishing returns from noise averaging at longer times.

$F_{assignment}$ rises steeply initially, reaching approximately 0.85 by 5000 ns before gradually plateauing toward unity. This rapid initial rise reflects the erf function's sensitivity to SNR in the intermediate regime: small improvements in SNR translate to substantial fidelity gains when distinguishing overlapping Gaussian distributions. The curve approaches but never quite reaches perfect fidelity because residual noise from the measurement chain always maintains some finite overlap between the $|g\rangle$ and $|e\rangle$ pointer states.

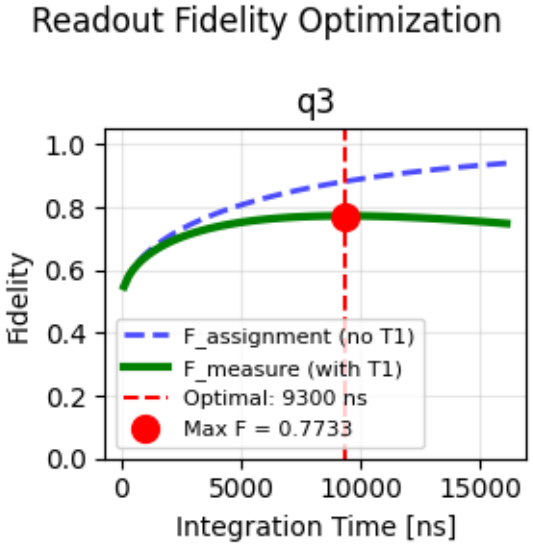


FIG. 3. Readout fidelity optimization for qubit q3 with $T_1 = 70.504\ \mu s$. Blue dashed line shows assignment fidelity (noise-limited only), green solid line shows total measurement fidelity including T_1 decay. The optimal integration time of 9300 ns (red vertical line) yields maximum fidelity F = 0.7733 (red dot), representing the optimal balance between SNR accumulation and qubit relaxation.

The green solid curve representing $F_{measure}$ tells a more complex story. Initially, it follows the assignment fidelity closely, rising rapidly as SNR improves. However, around 5000 ns, the exponential decay factor e^{-t/T_1} begins to dominate, causing the measurement fidelity to peak and then decline. The optimal timing was reached at approximately 9300 ns, where $F_{measure} = 0.7733$. This represents only 10.2% of the qubit's T_1 lifetime. This early optimization point suggests that our readout achieves sufficient SNR relatively quickly, such that the marginal benefit of continued integration does not outweigh the accumulating decay probability.

Figure 4 provides crucial diagnostic information by decomposing the measurement into its constituent components: pointer separation $|\Delta\alpha|$ (blue, left axis) and noise standard deviation σ (orange, right axis). The blue curve shows the integrated signal growing linearly with time, reaching approximately 0.45 mV by 15000 ns. This linear scaling ($\Delta\alpha \propto t$) confirms that the resonator has reached its dispersive steady state early in the integration window and that our post-processing correctly simulates shorter integration times by scaling the measured values. The orange curve demonstrates the integrated noise growing as $\sqrt{t}$, increasing from near zero to approximately 0.125 mV by 15000 ns.The purpose of this diagnostic is to display the growth of both parameters over time following our theory.

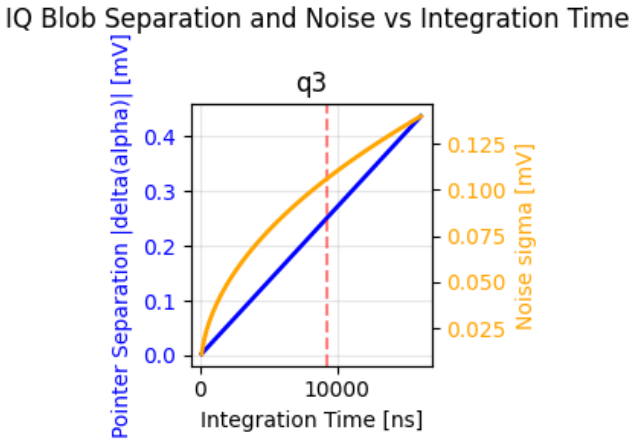


FIG. 4. Decomposition of SNR into signal and noise components. Blue line (left axis) shows pointer separation $|\Delta\alpha|$ growing linearly ($\propto t$) as the integrated signal accumulates. Orange line (right axis) shows noise σ growing sublinearly ($\propto \sqrt{t}$) due to statistical averaging. The diverging gap between these curves explains why SNR $\propto \sqrt{t}$. Red dashed line marks the optimal integration time. Note scales between mV are different, as they are made to reflect scaling factors instead of difference.

The diverging gap between signal and noise curves explains the physical origin of the SNR $\propto \sqrt{t}$ scaling: since signal grows faster than noise, their ratio necessarily increases with time. At the optimal time of 9300 ns (red dashed line), the signal-to-noise ratio has improved suffi-

ciently that assignment fidelity is reasonable, but not so long that T_1 decay has significantly eroded the measurement fidelity. The absolute magnitudes signal separation of ∼0.3 mV and noise of ∼0.1 mV at the optimum suggest moderate dispersive coupling and reasonable amplifier noise performance, though both could potentially be improved for higher peak fidelity.

The experimental validation of the $\sqrt{t}$ SNR scaling and the clear identification of an optimal integration time demonstrate that our QUA node successfully implements the theoretical framework described in Section I. The results confirm that dispersive qubit readout fundamentally involves a trade-off between two competing time scales: the measurement time scale (set by resonator linewidth κ and SNR accumulation rate) and the decoherence time scale (set by T_1).

IV. DISCUSSION

A. Comparison to Baseline

Notably, this optimal value of 9300 ns is substantially shorter than the baseline Qolab setting of 16000 ns (4000 QUA time units). The optimization therefore successfully identified a sweet spot that improves readout latency by 42% while maintaining the best achievable fidelity given our current hardware parameters. The ratio $t_{\text{opt}}/T_1 \approx 0.1$ is unusually low compared to typical dispersive readout implementations. This typically indicates that either (a) the SNR coefficient $|\Delta\alpha_{\text{ss}}|/\sigma$ is already quite good, allowing high fidelity at short times, or (b) there are technical limitations preventing us from realizing the full potential of our qubit. Given that our peak fidelity is only 77%, the latter explanation is more likely: improving the readout chain (resonator coupling, drive amplitude, parametric amplifier gain, or mixer calibration) could increase the SNR coefficient, shifting the optimal time to longer values and achieving higher peak fidelity closer to the 95-99% range expected for well-optimized systems [2].

The relatively early optimal time suggests we are operating in a regime where T_1 decay becomes significant before achieving high assignment fidelity. With the qubit $T_1 = 70.504$ μs the readout should theoretically be able to integrate much longer if the SNR coefficient were larger, corresponding to less variance in measured IQ distributions or larger discrimination between state amplitudes. However, shorter integration times are preferable in practice for quantum computing applications. Modern error correction protocols such as the surface code require rapid, repeated measurements to track error syndromes in real time [5]. Integration times in the range of hundreds of nanoseconds to a few microseconds enable higher measurement throughput, which directly translates to faster quantum algorithms and more efficient error correction cycles. Therefore, while our hardware could support longer integration for marginally higher fidelity, the 9.3 μs result represents a practical compromise prioritizing computational speed.

B. Limitations and Systematic Uncertainties

The peak fidelity of 77.33% is suboptimal for fault-tolerant quantum computation, which typically requires readout fidelities exceeding 99% [4]. This indicates several opportunities for improvement in our experimental setup. First, the baseline SNR reflected in the steady-state pointer separation $|\Delta\alpha_{\text{ss}}|$ and single-shot noise σ could be enhanced through multiple hardware optimizations. Increasing the readout drive power would produce larger steady-state amplitudes α_g and α_e, thereby increasing their separation $|\Delta\alpha_{\text{ss}}|$. Improved parametric amplification represents another avenue: quantum-limited Josephson parametric amplifiers can approach the standard quantum limit for phase-preserving amplification, substantially reducing the noise temperature of the measurement chain [6]. Better mixer calibration to reduce IQ crosstalk and minimize DC offsets would also directly improve the Gaussian separation between pointer states. While all parameters were optimized prior to running this experiment using standard Qolab calibration modules, the inherent drift and noise in superconducting systems means these calibrations are never perfect.

A significant practical limitation is the temporal variability of T_1. We observed that the T_1 time is a dynamic feature, fluctuating between measurement runs due to two-level system (TLS) defects in the substrate and interfaces, fluctuations in the magnetic flux environment, and variations in residual quasiparticle density [7]. This variability has direct implications for our optimization: the formula in Equation (5) utilizes a fixed T_1 value measured at a specific time, but the actual decay rate during readout optimization may differ. If T_1 was shorter during the readout sweep than during the T_1 calibration run, the true optimal integration time would be shorter than calculated. Conversely, if T_1 improved, we would be integrating less than optimally. This temporal instability likely explains why the baseline Qolab state file specifies a substantially longer integration time of 16000 ns: a conservative default that remains safe even when T_1 degrades. A more sophisticated approach would involve real-time T_1 tracking or periodic recalibration to adapt the readout integration time dynamically.

There are also fundamental limitations in our theoretical model. The decay fidelity factor e^{-t/T_1} in Equation (5) is an approximate model that assumes exponential relaxation with a single time constant. In reality, qubit relaxation can exhibit non-exponential behavior due to multiple decay channels, Purcell filtering effects that modify the effective T_1 during readout, or quantum jumps that occur stochastically rather than deterministically [1]. Additionally, our model treats assignment fidelity and decay fidelity as independent and multiplicative, which is a simplification. In practice, these effects

can couple: for example, if the qubit jumps from $|e\rangle$ to $|g\rangle$ early in the integration window, the measurement will accumulate mostly ground-state signal, whereas a late jump will accumulate mostly excited-state signal. This creates a continuous spectrum of outcomes rather than discrete "decayed" or "not decayed" events, leading to more complex error statistics than our simple model captures. Nevertheless, the phenomenological model provides an excellent first-order approximation and successfully identifies a near-optimal integration time.

We also attempted to implement a confusion-matrix-based comparison to display the fidelity before and after the qubit readout optimization, however the qubits lost stability after the last test implementation and we had to go forward with the last available data. This is also the reason for the ≈ 70 μs T1 time, since future attempts saw the qubits with worse or less confident results.

C. Next Steps and Future Work

Beyond simply identifying the single optimal integration time, a natural extension of this work would be to explore the fidelity-latency trade-off space more systematically. Rather than strictly maximizing fidelity, many quantum computing applications could benefit from a family of readout protocols optimized for different operating points. For instance, we could define a tolerance threshold, say accepting any fidelity within 95% of the maximum achievable fidelity, and then minimize the integration time subject to that constraint. Figure 3 shows that fidelity within a few percent of the peak can be achieved with significantly shorter integration times: the curve is relatively flat near the optimum, meaning that reducing integration time from 9300 ns to perhaps 6000 ns might only cost 1-2% in fidelity while providing a 35% reduction in latency. This would enable a tunable parameter where users could select "fast mode" (lower fidelity, 5-6 μs) versus "high-fidelity mode" (77% fidelity, 9.3 μs) depending on the requirements of their algorithm.

A more sophisticated optimization could incorporate the specific error budget of the target application. For example, quantum error correction codes have different sensitivities to bit-flip versus phase-flip errors, and readout errors contribute differently to these error channels compared to gate errors [5]. An application-aware optimization might accept lower readout fidelity if it enables faster syndrome extraction cycles that reduce the accumulation of other errors. Additionally, machine learning techniques could be employed to optimize readout in real time: an online learning algorithm could continuously adjust integration time based on measured qubit coherence, drifting optimal frequency, or observed readout fidelity, creating an adaptive calibration system that responds to environmental changes.

From a hardware perspective, future work should focus on improving the fundamental SNR coefficient $|\Delta\alpha_{\text{ss}}|/\sigma$ rather than simply optimizing integration time. Increasing the dispersive shift 2χ through improved circuit design, implementing Purcell filtering to suppress qubit relaxation during readout, upgrading to quantum-limited parametric amplification, and employing optimal control techniques to shape the readout pulse could collectively push fidelities toward the 99%+ regime required for fault-tolerant quantum computation [4].

Finally, extending this optimization framework to simultaneous multiplexed readout, or multi-qubit readout, introduces additional complexity. Cross-talk between readout resonators, Purcell decay induced by neighboring qubits, and amplifier saturation effects can all modify the optimal integration time in non-obvious ways [2]. A comprehensive calibration node that jointly optimizes integration times for all qubits in a multi-qubit processor, accounting for these interaction effects, represents an important direction for scaling this work to larger quantum systems.

V. CODE

Find the Node in GitHub here

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